INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

Ute Hahn
October 23, 2026

Statistical Models
Week 9: Objective priors and numeric methods

Topics for this week include:

- More on priors: examples for conjugate priors, mixtures of conjugate priors, and objective priors, in particular *Jeffreys' prior*.
- Calculating summaries for the posterior when the prior is not the conjugate prior: calculation of maximum a posteriori estimate (MAP), numerical integration, and Laplace's approximation for the posterior: the "Bayesian CLT".
- If time allows, we will discuss the general idea of Markov Chain Monte Carlo methods. These methods are based on results for Markov chains – we will shortly review these results. You may know Markov chains with finitely many possible outcomes, that is, the values are integers (finite state space). Here, we use Markov chains with real valued outcomes (continuous state space).

Relevant Readings

- More on priors: R&G, Sections 2.1, 2.3.1
- Numerical methods: R&G, Section 3.1
- Notes on Markov chains with finite and continuous state space

Lecture Material

- Link to onenotes on Google drive
- Code for R&G, Figure 2.5
- Code for R&G, Figure 3.2
- (evt.) Experiments with AR(1) chains

Exercises 9

Problem 9.1: Negative binomial – beta model

Reich & Ghosh, Chapter 2, Exercise 4.
Assume that $Y \mid \theta \sim \text{NegBinomial}(\theta, m)$ (see also Appendix A.1 in Reich and Ghosh) and $\theta \sim \text{Beta}(a, b)$.

a. Derive the posterior of θ.
b. Plot the posterior of θ and give its 95% credible interval assuming $m = 5, Y = 10$, and $a = b = 1$.

Problem 9.2: Modelling side effects of medication

Reich & Ghosh, Chapter 2, Exercise 8.
The Mayo Clinic conducted a study of $n = 50$ patients followed for one year on a new medication and found that 30 patients experienced no adverse side effects (ASE), 12 experienced one ASE, 6 experienced two ASEs and 2 experienced ten ASEs.

a. Derive the posterior of λ for the Poisson/gamma model $Y_1, \dots, Y_n \mid \lambda \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
b. Use the Poisson/gamma model with $a = b = 0.01$ to study the rate of adverse events. Plot the posterior and give the posterior mean and 95% credible interval.

c. Repeat this analysis with Gamma$(0.1, 0.1)$ and Gamma$(1, 1)$ priors and discuss sensitivity to the prior.
d. Plot the data versus the Poisson$(\hat{\lambda})$ PMF, where $\hat{\lambda}$ is the posterior mean of λ from part (b). Does the Poisson likelihood fit well?
e. The current medication is thought to have around one adverse side effect per year. What is the posterior probability that this new medication has a higher side effect rate than the previous mediation? Are the results sensitive to the prior?

Problem 9.3: Priors and parametrization

In this exercise, you look at different possibilities to parametrize a prior distribution, and at the consequences: what does a prior on one parametrization mean for the other parametrization? Additionally to (or instead of) simulations, you could also try to calculate the induced prior densities and plot them over the histograms.
Hint: You do not need to do any Bayesian analysis in this exercise, just look at the distributions of γ (in parts a and b) and of θ (in c and d), respectively.
Here is the exercise (Reich & Ghosh, Chapter 2, Exercise 11):
Let θ be the binomial success probability and $\gamma = \theta/(1-\theta)$ be the corresponding odds of a success. Use Monte Carlo simulation (you can discard samples with $\gamma > 100$) to explore the effects of parameterization on prior choice in this context.
a. If $\theta \sim$ Uniform$(0, 1)$, what is the induced prior on γ?
b. If $\theta \sim$ Beta$(0.5, 0.5)$, what is the induced prior on ?
c. If $\gamma \sim$ Uniform$(0, 100)$, what is the induced prior on θ?
d. If $\gamma \sim$ Gamma$(1, 1)$, what is the induced prior on θ?
e. Would you say any of these priors are simultaneously uninformative for both θ and γ?

Problem 9.4: Proper and improper priors

In this exercise, you examine two models that relate to the normal distribution. In a) we have a case where an improper prior leads to a proper posterior, wheras in b), where the model is a mixture of normal distributions, it is not possible to find any improper prior that still gives a proper posterior.
a. Say $Y \mid \mu \sim$ Normal$(\mu, 1)$ and μ has improper prior $\pi(\mu) = 1$ for all μ. Prove that the posterior distribution of μ is a proper PDF.
b. Consider a sample of size n from the mixture of two normal densities given by $f(y \mid \theta) = 0.5\phi(y) + 0.5\phi(y - \theta)$ where $\phi(z)$ denotes the standard normal density function. Show that any improper prior for θ leads to an improper posterior.

see Reich & Ghosh, Chapter 2, Exercises 12 and 14.

Problem 9.5: Conjugate and Jeffreys' priors for exponential models

Assume that $Y \mid \lambda \sim$ Exponential(λ), that is, $Y \mid \lambda \sim$ Gamma$(1, \lambda)$, where λ is the rate parameter.
a. Find a *conjugate prior* distribution for λ and derive the resulting posterior distribution.
b. Now consider the *Jeffreys' prior* for λ.
   i. Derive and plot the Jeffreys' prior for λ.
   ii. Is this prior proper?
   iii. Derive the posterior and give conditions on Y to ensure it is proper.

compare Reich & Ghosh, Chapter 2, Exercises 3 and 16.